

Urban Polycentricity and the Costs of Commuting: Evidence from Italian Metropolitan Areas

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ABSTRACT Polycentricity at the metropolitan scale is perhaps the model of spatial organisation that needs to be investigated more thoroughly as regards its effects on travel. The aim of this paper is to test the role of polycentricity—as well as other spatial characteristics, such as compactness, functional diversification and size—in the costs of commuting, taking into account an external cost component (per-capita CO₂ emissions) and a private cost component (time spent on travelling). The degree of urban polycentricity has been measured by adopting a dynamic approach based on commuting flows and on social network analysis tools. The analysis is carried out using a database of 82 Italian metropolitan areas (MAs). Results show that MAs with a higher degree of polycentricity are more virtuous both in terms of private and external costs of mobility, while the degree of compactness is associated with lower environmental costs but with higher private costs. Size is associated with both higher external and private costs, while the role of functional diversification turns out to be statistically insignificant. Socio-demographics also play a role.

Introduction

The spatial organisation of cities and its effects on welfare, economic growth, and environmental sustainability has increasingly captured the attention of economists and other social scientists in the past two decades. In this debate, one of the most relevant topics is about the relationship between the characteristics of urban spatial organisation—i.e., polycentricity, dispersion, size, and mixed land use—and mobility patterns (Ewing and Cervero 2001). Mobility yields relevant social costs that include both private components (i.e., bus ticket, fuel and time spent travelling) and external components (i.e., negative environmental externalities like global warming and pollution). While the former directly affects the

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commuters' utility, the latter affects the welfare of the whole society and it is not considered in the individual's travel decisions.

Among the spatial characteristics of cities, the degree of polycentricity is perhaps the one for which most effort must still be made to shed some light on its economic effects and externalities, especially those generated by travel behaviour (Schwanen, Dijst, and Dieleman 2002). The relevance of polycentricity arises from the fact that polycentric spatial organisation has been playing a relevant normative role for more than 10 years. In fact, polycentricity is now considered a planning tool to enhance cities' competitiveness, social cohesion, and environmental sustainability (Davoudi 2003). However, these hypotheses have not yet been empirically investigated enough. Moreover, polycentricity itself needs to be better defined and measured, especially at the metropolitan level. Most of the attempts to measure urban polycentricity in the literature are based on static and morphological approaches, which mainly consider the spatial distribution of employment or of population within metropolitan areas (MAs), without focusing on the interactions among urban nodes.

The aim of this paper is to test whether a high degree of urban polycentricity is virtuous in terms of the costs of mobility, and thus, can be considered as a desirable planning goal. While trying to answer this question, other relevant characteristics of urban spatial organisation have also been considered as potential determinants of the commuting costs. More in depth, the role of urban size, dispersion, and functional diversity have been assessed.

The analysis of polycentricity at a metropolitan level requires a definition and identification of sub-centres. From a morphological perspective, urban sub-centres are usually identified with employment density thresholds. In this paper, an attempt is made to identify sub-centres after having defined them from an interaction perspective rather than from a morphological one. In this way, urban sub-centres are defined as places with central functions that organise the territory around them. Their identification is hence carried out using dynamic approaches based on social network analysis tools, which allow to capture the centrality of urban nodes and their hierarchic ranking.

In section 2, a review of the literature on the relationship between urban structure and mobility patterns is carried out, with particular attention on the role of polycentricity. Section 3 defines the concept of polycentricity and proposes two indicators—for reasons of robustness—to measure it appropriately. Section 4 reviews the commuting patterns in the Italian MAs and proposes two indicators to measure external and private costs related to mobility, respectively. The former has been measured by focusing on CO₂ emissions, while the latter has been approximated by the time spent on travelling. In section 5, the causal links

between the characteristics of urban spatial structure and costs of mobility are explained and then tested empirically with an econometric cross-section analysis. Section 6 concludes.

Literature: A Still Open Debate

The role of urban structure is now recognised as a factor that facilitates travel because it provides the framework for the location of houses and firms (Hickman and Banister 2007). The characteristics of spatial organization that are supposed to be the most relevant in shaping commuting patterns at a metropolitan level are basically polycentricity, dispersion, mixing use and size (Tsai 2001).

Polycentricity and commuting. Polycentricity is perhaps the characteristic on which knowledge is less consolidated among scholars, regarding its effects on commuting patterns (Aguilera 2005; Levine 1992; Schwanen, Dijst, and Dieleman 2002). Some authors (Gordon and Wong 1985; Tsai 2001) argue that polycentricity has virtuous effects on mobility patterns. This has been put forward for several reasons.

First, polycentric metropolitan organisation favours proximity between work and home—the so-called “co-location hypothesis” (Gordon and Wong 1985: 666), reducing the distance travelled (Handy 1996). Distances can be reduced also because of lower rent near sub-centres if compared with monocentric areas (Cervero and Wu 1997), encouraging housing near the workplace. However, the rigidity in residence mobility—especially in the short run—can in part compensate for these effects.

Second, polycentric areas can favour the competitiveness of mass transit (Breheny 1995; Susilo and Maat 2007), reducing the use of less sustainable private means of transport (e.g., cars, motorbikes, etc.). In fact, sub-centres ensure a minimum dimensional scale as regards the supply of a competitive public transport system. Moreover, polycentric MAs, which are characterised by two or more central nodes, are the natural environment for cross-commuting patterns between such nodes, which can favour the use of the public transport system (Susilo and Maat: 607).

Third, polycentricity can also influence the time spent on travelling. Polycentric urban structures are thought to reduce the excessive congestion that characterises monocentric areas, especially those of big dimension (Parr 2004: 235–236). Diseconomies of congestion can be avoided through the distribution of population, employment, and urban functions in several sub-centres, without renouncing the advantages of agglomeration. In other words, polycentricity can be interpreted as a fair compromise between the need for safe urban compactness and the need to lighten congestion, while the city is growing.

Despite the above-mentioned arguments that make polycentricity a potentially desirable planning goal, only a few empirical analyses have been carried out to corroborate these hypotheses. What is more, the available empirical works show quite contrasting results, so that the relation between polycentricity and the costs of mobility still remains unclear. In a work on the MA of San Francisco, for example, Cervero and Wu (1998) note that decentralisation of employment in several sub-centres is associated with an increase in commuting distances and in car drivers. Other scholars have found similar results, even for European cases (Aguilera 2005; Aguilera and Mignot 2004; Levinson and Kumar 1994; Næss 2007). Schwanen, Dieleman, and Dijst (2004), in an analysis of the Dutch urban system, show that polycentric cities—if compared with monocentric ones—are associated with longer commutes, both in terms of time and distance.

On the other hand, other studies show that decentralisation of employment tends to reduce commute duration, both in Europe (Alpkokin et al. 2005; Owens 1986) and in the U.S. (Gordon, Richardson, and Jung 1991; Tsai 2001). Contrasting results can be found even with respect to the share of mass transit users. In particular, several studies show that the progressive transformation of cities from monocentric to polycentric is associated with a decrease in public transport users (Cervero and Wu 1998; Næss 2007; Schwanen, Dijst, and Dieleman 2002), while others show the contrary (Bento et al. 2005).

These contrasting results are at least partially due to the way in which the concept of polycentricity is defined and measured. In almost all the cases, polycentricity is intended to be the result of a mere process of decentralisation from the CBD to the emergent sub-centres. From this perspective, a sub-centre is simply defined as a place with an employment density that is higher than the surrounding territory. This interpretation can be sound for North-American cities, while it is less appropriate for European ones, where urban areas formed mostly through a process of coalescence rather than decentralisation (García López and Muñiz 2005). As a consequence, the analysis of urban polycentricity in European cities must take into account a more complex definition of urban centres, which can be interpreted as high-hierarchy places that supply central functions to the surrounding territory. Hence, the simple density-based approach does not seem to be correct.

In addition, the fact that in the last 20 years an important phenomenon of decentralisation has occurred, in tandem with an increase of commuting distance travelled, does not necessarily mean that polycentricity yields worse commuting patterns. Once again, the process of decentralisation that occurred in almost all the European MAs has not necessarily led to polycentricity. This process, in fact, is

associated with a twofold evolution: polycentric development or simply dispersed urban areas (Lee 2007). This could be another reason to explain the ambiguity of the results of empirical works.

The role of the other spatial characteristics. In addition to polycentricity, other spatial characteristics of cities can shape commuting patterns and their effects. First, urban dispersion is considered one of the most important spatial determinants concerning the increase in commuting distances (Banister 1999), mass transit inefficiency (Ellison 1995) and energy consumption (Næss 1995; Newman and Kenworthy 1989). There is already quite a wide consensus among scholars over the disequilibria and externalities—especially the environmental ones—generated by low density settlements and scattered patterns of urban development, even for Italian cities (Calafati 2003). On the other hand, the effects on private costs, like the time spent on travelling, are still unclear. Regarding U.S. cities, for example, Glaeser and Kahn (2004) and Gordon, Kumar, and Richardson (1989) observe that higher densities are associated with higher travel times. On the other hand, Levinson and Kumar (1997) find a negative relationship between density and travel times when density is beyond a given threshold, after which the sign changes.

Functional diversification represents another characteristic of urban spatial organisation that can be important in shaping commuting patterns within cities. An urban area is functionally diversified if it presents a job-housing balance or, more generally, a diverse or mixed land-use (Camagni, Gibelli, and Rigamonti 2002; Cervero 1996). Jane Jacobs (1961) was one of the major advocates of such a model of spatial organisation, even for cities' economic prosperity and safety. The empirical literature shows that the higher the level of diversity, the shorter the distance travelled (Cervero 1996) and the higher the transit usage (Frank and Pivo 1994). In an analysis of six Italian MAs, Travisi and Camagni (2005) show that the collective costs of mobility decrease in areas with higher density and mixed land-use.

Another key factor in shaping commuting patterns is the generalised deregulation of house building that occurred in Europe during the last 20 years. This, together with changing individual preferences and the widespread diffusion of private transport, combined to bring about less transit oriented cities and longer commuting travels. Newly built areas are often planned without taking enough into account the needs for an efficient mass transit service, yielding scattered and high land consumption settlements. Hence, the rapidity of the house building process seems to be associated with less sustainable mobility patterns (Camagni, Gibelli, and Rigamonti 2002). Cirilli and Veneri (2007), in an analysis of the major Italian Local Labour Systems, show that

a newer housing stock is associated with higher polluting emissions due to commuting.

Urban Polycentricity: Concept and Measure

Polycentricity is emerging as an archetypal model of spatial organisation of MAs (Anas, Arnott, and Small 1998). Polycentric areas can be conceptualised based on two main dichotomies, depending on the perspective adopted. On the one hand, adopting a pure morphological approach, polycentric areas can be viewed as a model of spatial organisation midway between traditional compact cities and urban sprawl. They maintain the advantages related to compact cities, complying with the spontaneous tendencies to dispersion (Camagni, Gibelli, and Rigamonti 2002: 52). On the other hand, adopting a both functional and morphological approach, polycentric areas represent an alternative to monocentric areas (Meijers and Sandberg 2008). Hence, polycentricity is considered as the natural evolution of monocentric spatial organisation. However, polycentricity can be the result of two different territorial evolutions. The first consists in the decentralisation processes from the Central Business District (CBD) to new emerging sub-centres. This pattern took place with particular intensity in U.S. cities. The second, which prevails in European cities, consists of a progressive integration of pre-existing self-contained centres in a single MA, through a coalescence (Calafati 2002) or an incorporation (Champion 2001) process.

Using a traditional morphological approach, an urban area can be defined polycentric if its employment is not concentrated in a single centre (Riguelle, Thomas, and Verhetsel 2007: 195), but it is distributed in two or more centres. Adopting a functional approach, on the other hand, the focus is not only on the distribution of employment, but—more generally—on the spatial distribution of functions (Hall 2009). With the functional approach, territorial units within an MA must show two or more nodes or “centres” with a higher hierarchical level. Given that, the definition of polycentricity at the urban level appears quite simple and should not be a matter of debate. Instead, the concept of sub-centre needs more attention. While from a pure morphological perspective a centre is simply a place with a higher employment density than the hinterland, from a wider point of view a centre is a place that carries out central functions and that organises the surrounding territory. In order to get hold of these features, it is possible to look at the way in which each node interacts with the others. Connections among nodes and their distribution tell a lot about the way in which each place organises its surrounding territory and about its hierarchic ranking within the MA.

The first step to measure the degree of urban polycentricity consists of the identification of sub-centres. For this purpose, there are a lot of studies that use

density-based approaches, especially in U.S. literature. Some of them use simple absolute cut-offs of employment density and of total employment for census tracts within MAs (Anderson and Bogart 2001; Cervero and Wu 1998; Giuliano and Small 1991). Other more sophisticated studies use relative density-based approaches, usually through the estimation of the local peaks of a density function, which often has a non-parametric specification (Craig and Ng 2001; McDonald and Prather, 1994).

Another family of approaches for the identification of urban sub-centres—which has been followed in this work—is based on “dynamic” indicators, which are built using flow data, such as commuting flows (Bourne 1989). Literature on these dynamic methodologies has been developing noticeably in the last decade. In a study of the Spanish MAs, Burns, Boix, and Roca (2001) use a threshold of net entry flow of commuters (15 percent) above which an area can be thought of as a sub-centre. They also add an absolute population dimension cut-off of 10,000 residents, similar to Giuliano and Small’s approach (1991). Other similar interaction approaches at the intra-urban level are those by Clark and Kuijpers-Linde (1994) and by Van der Laan (1998). The latter study does not identify sub-centres, but it divides a set of urban areas into four types of spatial structure looking at patterns of commuting flows.

Other dynamic approaches have been used, especially at the inter-urban level, by the GaWC (Globalization and World Cities) research network (Derudder et al. 2003; Taylor et al. 2002). These approaches start conceptualising the network of cities from a micro-founded perspective, looking at the location strategies of global service firms and at the flows they generate. Derudder and Witlox (2008) give another contribution to the investigation of the spatiality of World City Network using airline flows. The work of Limtanakool, Dijst, and Schwanen (2007) represents another step ahead towards the identification and characterisation of urban spatial structure from an interaction perspective. The authors consider three dimensions, which they call the “three S” for characterising polycentricity at the inter-urban level. The first two are the strength and the symmetry of the interactions, while the third one is the structure of the network. The identification of centres is carried out through the use of some dominance indexes.

In this work, sub-centres of Italian MAs have been identified using a social network analysis approach. The methodology starts conceptualising MAs as networks of municipalities—composed of nodes (municipalities) and links (flows of commuting)—and then it uses the in-degree index to characterise the degree of centrality of each node, as explained in Boix (2002). More in depth, for each municipality within each MA, the in-degree index has been computed using commuting flows data in 2001. The in-degree index measures the number of links

that go towards a given node (municipality) and it represents a simple, direct and stable measure of centrality among a given network's set of nodes (Costembader and Valente 2003: 291; Zemljič and Hlebec 2005: 82). Moreover, Alderson and Beckfield (2004) see the incoming flows also as an important index of a city's prestige. In-degree has been calculated using dichotomised four nearest neighbour matrixes of commuting,¹ applying the following formula:

$$I_j = \sum_{i=1}^N z_{ij} \quad (1)$$

where I_j is the in-degree indicator for the municipality j of a given metropolitan area; z_{ij} is the link between municipality i and municipality j , which is 0 or 1, depending on whether i is one of the four nearest neighbours of j . Using a k -nearest-neighbour specification of the matrixes, the in-degree also takes into account—albeit indirectly—the size of the flows.

Once having ranked all the municipalities on the basis of the value of the in-degree index, those belonging to the 95th percentile were selected and considered as sub-centres. In this way, all municipalities that show high attractiveness in terms of labour flows can be considered as “central places,” without considering any density-based measure. Then, the degree of urban polycentricity—taking into account the size differences among MAs—was quantified on the basis of the assumption that the higher the number of sub-centres in a given MA, the higher its level of polycentricity.

Hence, a scatter-plot between the number of identified sub-centres and population was drawn (Figure 1), together with a line that interpolates the two variables. Each point in the graph represents an MA with its number of sub-centres and residents. The vertical distance between each point and the line—or, in other words, the residual of the interpolation—can be interpreted as a direct measure of an MA's degree of polycentricity.

In order to add robustness to the measurement of polycentricity, another indicator was applied—called Special Functional Polycentricity (P_{SF}) (Green 2007). Again, this index employs a dynamic approach based on tools of social network analysis and it is computed using commuting flows, with a network-level perspective. The P_{SF} index—which ranges from 0 to 1—can be calculated as follows:

$$P_{SF}(N) = \left(1 - \frac{\sigma_\partial}{\sigma_{\partial\max}}\right) \cdot \Delta \quad (2)$$

where σ_∂ is the standard deviation of nodal in-degree within the MA N ; $\sigma_{\partial\max}$ is the standard deviation of the nodal in-degree of a two-node network (n_1, n_2) derived from N where $d_{n1} = 0$ and $d_{n2} =$ value of the node with the highest value in N ; Δ is

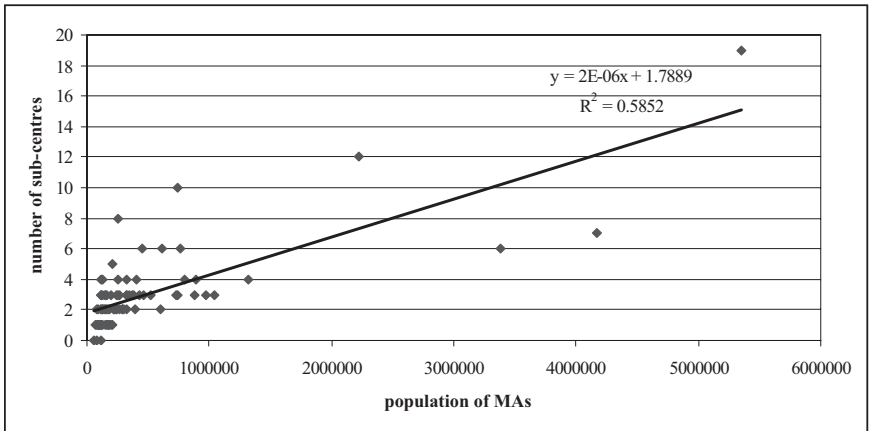


FIGURE 1. SCATTER PLOT WITH MAS' POPULATION AND NUMBER OF SUB-CENTRES.

the density of the network (Green 2007: 2085).² Higher values indicate a higher degree of functional polycentricity. Following this indicator, Figure 2 represents Italian MAs on the base of their degree of polycentricity.

The main merit of the P_{SF} is that it concentrates in a single measure many features that characterise polycentricity, which in other studies is quantified using a set of indexes, as in Limtanakool, Dijst, and Schwanen (2007). Indeed, the P_{SF} index takes into account some features that have not been directly considered in the previous measure based on the number of sub-centres because it looks at the same phenomenon from another angle. The P_{SF} index considers the degree of hierarchy of the network through the first multiplicative factor and, at the same time, it considers the strength of the interaction by including the density of the network.

Commuting Patterns in the Italian Metropolitan Areas

Work commutes in Italy. Before analysing the relationship between spatial characteristics and commuting costs, it is worth exploring the patterns of mobility in Italy through simple descriptive statistics. The units of analysis are 82 Italian MAs, which are identified applying the functional algorithm based on the Federal Register's SMA methodology (Office of Management and Budget 1990), with the adaptations of Clusa and Roca (1997).³

Within the 82 MAs that have been considered in this analysis, a large number of people (almost 11 million) travel every day for work. As regards the mode

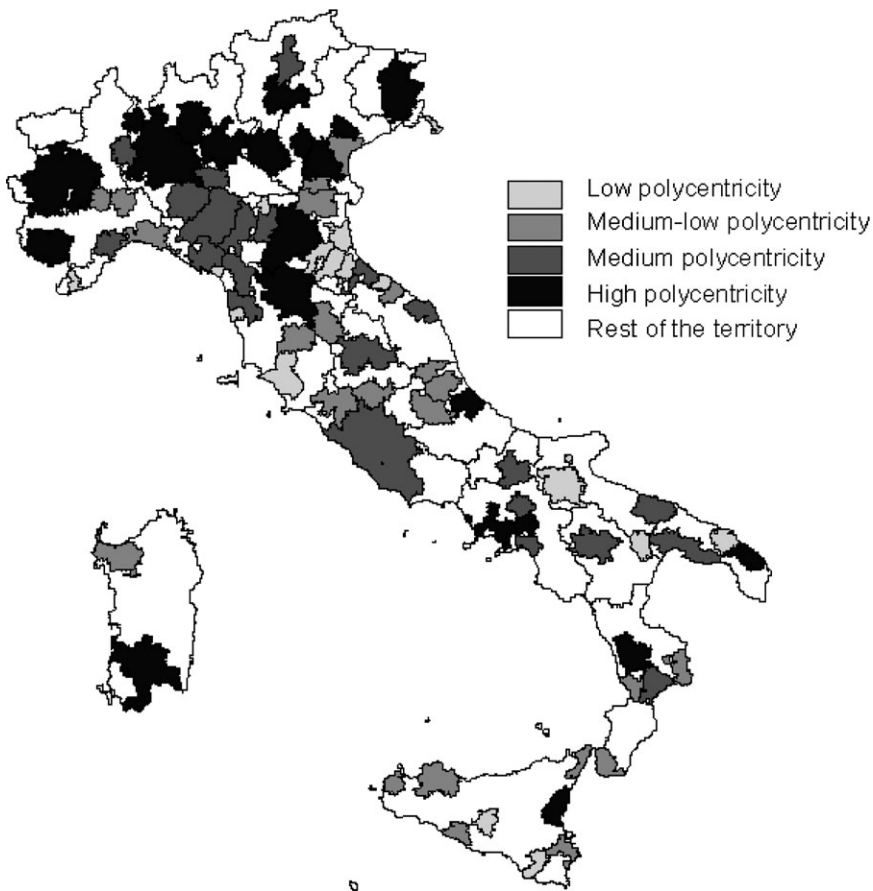


FIGURE 2. ITALIAN MAS AND THEIR DEGREE OF POLYCENTRICITY: THE P_{SF} INDEX.

choice, Table 1 shows that the most used means of transport is the private car (68.27 percent). On the whole, private means of transport are much more used than public means. In the latter, buses play a major role, followed by trains.

On average, distances travelled by public means of transport are longer than those by private means (Table 2). Moreover, it appears that the share of mass transit users increases with average distances travelled. In other words, it seems that public transport becomes convenient when the distance to be travelled is longer.

Another point to highlight is that, on average, travel duration with private means of transport is much shorter when compared to that with public transport

TABLE 1. MODES OF COMMUTING WITHIN ITALIAN MAS. ABSOLUTE AND PERCENTAGE VALUES.

Mode	Mode description	Abs. value	% value
1	Train	184,137	1.68
2	Tram	80,684	0.74
3	Underground	170,011	1.55
4	Urban bus or trolley bus	461,937	4.22
5	Extra urban bus or coach	153,504	1.40
6	School or business bus	67,744	0.62
	<i>Public transport</i>	<i>1,118,017</i>	<i>10.23</i>
7	Private car (driver)	6,952,851	63.59
8	Private car (passenger)	511,400	4.68
9	Motorbikes or scooter	689,287	6.30
10	Bike, foot or other means of transport	1,662,219	15.20
	<i>Private transport</i>	<i>9,815,757</i>	<i>89.77</i>
	<i>Total transport</i>	<i>10,933,774</i>	<i>100.00</i>

Source: Our elaboration on Istat Population Census data (2001).

(Table 3). This could be due to two different reasons. First, mass transit users commute longer distances, hence it takes more time for them to get to their destination. However, this factor alone can hardly explain all the differences in terms of time spent on travelling. Thus, another factor could be the minor time-efficiency of public means, probably due to the fact that it takes a relevant amount of time to reach the bus or train station from the traveller's home, as well as to reach the destination once off the train. The relative speed of private means can also play a role.

The private and external costs of commuting. This analysis relies on verifying whether polycentric spatial organisation influences the external and private costs of mobility within MAs. The external effects of mobility have been approximated in terms of CO₂ emissions, which have been measured through the Environmental Impact of Mobility (*EIM*) index, on the basis of mode choice and distance travelled by commuters. The *EIM* index has been calculated as follows:

$$EIM_h = \frac{\sum_{i,j,k} f_{i,j,k} w_k d_{ij}}{\sum_{i,j,k} f_{i,j,k}} \quad (3)$$

TABLE 2^a. DISTANCES TRAVELLED BY DURATION CLASS (IN MINUTES) AND BY MODE CLASS WITHIN MAS.

Time spent	Mode	Commuters ^c	%	Av. Km ^b
0–15	Private	4,104,296	44.3	10.4
	Public	119,007	1.3	13
15–30	Private	2,781,277	30.0	15.5
	Public	369,630	4.0	15.9
30–60	Private	1,122,150	12.1	22.7
	Public	474,610	5.1	21.4
>60	Private	145,815	1.6	27.9
	Public	154,770	1.7	31.1
		9,271,555	100.0	

^a The total number of commuters in Table 1 and in Table 2 differ because the latter Table does not consider commuters travelling on foot or by bike.

^b The distance are in km as the crow flies from a centroid to a centroid.

^c The total number of commuters is depurated from all the commuters for which the mean of transport or the time has not been releveled.

Source: Istat Census data (2001); Distances computed with GIS software.

TABLE 3. DISTANCES AND DURATIONS OF TRAVELS BY MODE CLASS WITHIN MAS.

Means of transport	Commuters	% commuters	Average duration (minutes)	Av. distance travelled (kilometres)
Public transport	1,118,017	10.2	38	10.4
Motorized private transport	8,153,538	74.6	19	8.3
Bike, on foot, other	1,662,219	15.2	10	2.6
Total	10,933,774	100	22	7.1

Source: Istat Census data (2001); Road distances computed with GIS software.

where $f_{i,j,k}$ is the number of commuters moving from municipality i to municipality j (within MA h), who use the k -th means of transport; w_k is the amount of CO₂ emissions (in grams) per passenger per kilometre, as estimated by Amici della Terra (2005) (see Table 4); finally, d_{ij} is the road distance between the two

TABLE 4. PER PASSENGER PER KILOMETRE
CO₂ EMISSIONS BY MEANS OF
TRANSPORT.

Means of transport	gr CO ₂ /pkm
Train	35
Tram	32
Underground	21.3
Urban bus	72
Extra-urban bus	26
School or company bus	31
Car	105
Motorbike	80
Bike, on foot, other	0

Source: Amici della Terra (2005).

municipalities. The *EIM* index gives a measure of the average quantity of greenhouse gases of the type-commuter in each Italian MA.

On the other hand, the private component of the costs of mobility has been approximated through the time spent on travelling per commuter. This component can be conceptualised as an internal cost that directly affects commuters' utility and their consequent behaviour in terms of mode choice. However, it could also be considered as a social cost because it is influenced by systemic inefficiency (Kapp 1950). Private commuting costs can include ticket and fuel prices as well as time spent on travelling. Given the limited availability of data, only the typical traveller's average commuting time (*ACT*) for each MA has been considered in this work. The *ACT* index has been calculated as in equation (4). Data for Commuting in 2001 was supplied by the Italian Institute of Statistics.

$$ACT_h = \frac{\sum_{i,j,k} f_{ijk} t_k}{\sum_{i,j} f_{ij}} \quad (4)$$

where $f_{i,j,k}$ is the number of commuters moving from municipality i to municipality j (within MA h), who spend time k on travelling; t_k is the duration class of the commute. Average value has been considered for each of the four classes, as shown in Table 5.

TABLE 5. TIME LENGTH OF COMMUTING
AND THEIR AVERAGE VALUE.

Time length classes (in minutes)	Time length considered (in minutes)
0–15	7.5
15–30	22.5
30–60	45
More than 60	75

Source: Elaboration on Istat Commuting data (2001).

The Spatial Determinants of Commuting Costs

The aim of this section is to test to what extent the characteristics of urban spatial structure influence the external (*EIM*) and private (*ACT*) costs of mobility. The hypothesis that has to be tested is that these commuting costs are a function of some characteristics of spatial structure which can be grouped in four categories.

The first one is the MAs’ degree of polycentricity. Sub-centres present a dimensional scale that facilitates the competitiveness of mass transit (Breheny 1995). Hence, a polycentric structure can facilitate a more intense use of public transport systems and this in turn negatively affects the external costs of mobility through a minor impact in terms of CO₂ emissions (Amici della Terra 2005).

Polycentricity, according to the co-location hypothesis (Gordon and Wong 1985), also increases job-house proximity. In fact, the presence of several sub-centres increases the probability of finding a job near the place of residence. This, in turn, allows a reduction in the distance and time spent on travelling, with a virtuous effect on commuting costs.

Urban compactness is also relevant in shaping commuting patterns and their effects. Compact cities, whose degree of compactness is usually measured with residential density, are expected to be associated with shorter travel distances, thanks to a higher spatial concentration of houses and job places. Moreover, urban compactness enhances the competitiveness of a mass transit system (Ellison 1995). On the other hand, the higher the cities’ compactness, the higher their level of traffic congestion, which in turn could be associated with an increase in commuting duration (Glaeser and Kahn 2004; Gordon, Kumar, and Richardson 1989).

Compactness can also be analysed in terms of spatial concentration of jobs within the metropolitan territory. For this purpose, the most applied measure is the Gini index between jobs and area shares of each municipality within each MA (Lee 2007). A higher concentration should be associated with shorter distances travelled, while the effects on travel duration could be ambiguous because of the congestion problems that may arise in highly concentrated areas.

Another variable of urban spatial structure that can influence commuting patterns is the functional diversification of the territory (mixed land use) (Camagni, Gibelli, and Rigamonti 2002). More specifically, a mixed land use favours the proximity between places of living and working. MAs with a high degree of functional diversification are generally characterised by an adequate equilibrium—within its territory—between residential and economic functions, integrating, rather than segregating them spatially. Thanks to a reduction in the distance between jobs and homes, functional diversification reduces the length of travel and, as a consequence, the emissions of CO₂ and of other pollutants. Functional diversification of the territory has been approximated with a Gini index between jobs and population shares within each MA (see Table 6). The more even the distribution of jobs and population within a given MA, the more diversified is its territory.

The last urban spatial characteristic that affects travel behaviour is city-size (Handy 1996). Total city surface area influences the average distance that has to be travelled, as well as the amount of time spent travelling. Moreover, big cities are expected to present an advantage in supplying efficient mass transit systems because it is possible to reach minimum economies of scale that are necessary to justify big investments and expensive infrastructures.

The logical relationship among the four main spatial characteristics and the dependent variables are visually represented in Figure 3, where left and right rectangles identify exogenous and dependent variables, respectively. Beyond the major spatial factors, a set of controls have been included in the analysis in order to take into account non-spatial determinants that can play a role in shaping commuting patterns. The role of these controls is not represented in the figure in order to keep it easy enough to be interpreted. The inclusion of such elements in the estimation, however, permits a better interpretation of the empirical results and helps avoid, or at least reduce, problems of under-specification of the model.

The first control is a measure of the average age of the housing-stock (*house_age*), which has been computed through the number of houses built after 1982 over the total number of houses. The underlying idea is that in the last decades the speed of house-construction has increased (Muñiz, López, and

TABLE 6. VARIABLES THAT HAVE BEEN USED IN THE ECONOMETRIC MODEL.

Variable name	Underlying spatial dimension	Description
polyc_green	Functional polycentricity	Special Functional Polycentricity index (Green 2007)
polyc_nsr	Functional polycentricity	“Relative sub-centre number” index of functional polycentricity (section 3)
gini_area_job	Spatial concentration of employment (morphological polycentricity)	Sum, for each municipality within a MA, of the differences in absolute value between the area and the employment shares of that municipality over the whole MA
ldensity	Urban compactness (inverse measure)	Log of the gross density (inhab./km ²)
gini_pop_job	Fuctional diversification—mixed land-use	Sum, for each municipality within a MA, of the differences in absolute value between the population and the employment shares of that municipality over the whole MA
house_age	Age of the housing stock	Number of houses built after 1982 over the total number of houses in 2001
rel_pub_comp	Mass transit competitiveness	Average km/h with mass transit over average km/h with private means of transport
area	Size	Total MA’s area (km ²)
dummy_centre	Geographic localization	MA’s of Central Italy
dummy_south	Geographic localization	MA’s of Southern Italy
demo_structure	Age of the population	Population structure index: people aged between 40 and 64 over people aged between 15 and 39
graduates	human capital	Share of people with a degree

Sources: Elaboration on Istat (1981; 2001a; 2001b) and Industry and Services Census (2001).

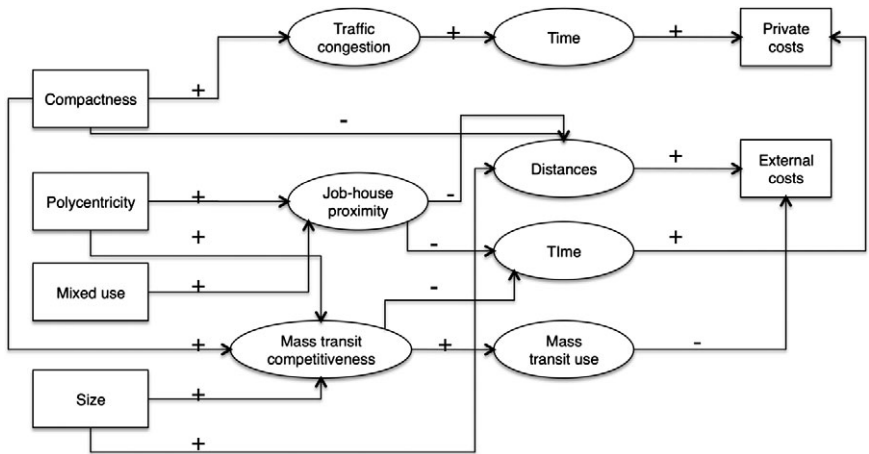


FIGURE 3. URBAN SPATIAL STRUCTURE AND COSTS OF MOBILITY: A QUALITATIVE MODEL.

Calatayud 2006), but new settlements have not been planned with enough attention to the proximity to mass transit service, in a general context of de-regulation (Camagni, Gibelli, and Rigamonti 2002). As a matter of fact, cities with a higher share of new houses are expected to be less efficient in terms of the external costs of commuting.

The relative competitiveness of mass transit service in comparison to private means of transport (*rel_public_comp*) can also play a role in shaping both private and external costs (see Table 6 to know how this simple indicator has been calculated). High mass transit competitiveness can boost the share of commuters who use these means of transport and can reduce the time spent travelling. Other controls that have been included in the analysis regard the geographical localisation of MAs within Italy, in order to take account of the large economic, cultural and technological differences among Italian macro-regions. The use of territorial dummies allows for these differences, although somewhat roughly.

Finally, two socio-demographic variables have been included in the model in order to control for different characteristics of the resident population. The first variable is the population structure index (*demo_structure*) (see Table 6), which reflects the age structure of the population, while the second variable is the share of graduates (*graduates*). A higher education level could be expected to be associated with a higher income and, as a consequence, a higher number of car owners that can influence commuting mode choice.

The empirical analysis. In order to test these hypotheses, a cross-section econometric model has been carried out using the above-mentioned 82 MAs as units of analysis. I acknowledge that, in order to establish causality, longitudinal data are to be preferred. However, given the limited availability of data, a cross-section has been employed. All the independent variables have been computed drawing on the Istat Population Census (1981, 2001) and the Istat Industry and Services Census (2001). Dependent variables have been computed as explained in section 4 (equations 3 and 4). Table 6 lists all the regressors, with a short description and measurement explanation. Regarding collinearity problems, they do not seem to arise, as shown by the correlation structure of the independent variables and their variance inflation factor (tables A1 and A2 in the Appendix).

As regards polycentricity, an additional variable of morphological distribution of employment within urban territory has been included (*gini_area_job*). More specifically, it has been calculated through the Gini index between the share of the area and employment of each municipality in each MA, as suggested in Tsai (2005) and Lee (2007). The higher the variable, the more concentrated the employment within each MA.

Two cross sectional econometric models have been estimated. The first regards external (environmental) costs (*EIM*), while the second adopts private costs (*ACT*) as dependent variable. The coefficients of the former model have been estimated through a spatial lag model, in order to clean the residuals from spatial dependence problems (see tables A3–A4 in the Appendix).⁴ This estimation is carried out with a maximum likelihood approach, weighting observation with a spatial matrix of contiguity. Model 2, on the other hand, has been estimated through an OLS method with robust standard errors.⁵ All the variables have been standardised with zero mean and unit variance so as to give a better interpretation of coefficients. Results are shown in Table 7.

As regards model 1, both the coefficients associated to the variables of functional polycentricity are negative and statistically significant. This result is confirmed by the coefficient relative to the *gini_area_job* variable, which tells us that the more concentrated the population within MAs, the lower the EIM indicator. These findings corroborate the hypothesis that a higher degree of polycentricity is associated with lower external costs of commuting in terms of average CO₂ emissions per traveller. The effect of polycentricity on the average time duration is also negative, but not statistically significant. Hence, it cannot be said that a higher degree of polycentricity makes the journey shorter. This result is in accordance with Levinson (1998), for whom the change in urban structure does not affect travel times. However, the coefficient of the *gini_area_job* variable tells us that the more concentrated the population, the longer the commutes.

TABLE 7. ESTIMATION RESULTS.

	Spatial lag models (ML)		Robust OLS	
dep. variable	EIM	EIM	ACT	ACT
polyc_green	-0.299*** 0.008		-0.092 0.341	
polyc_nsr		-0.184*** 0.006		-0.077 0.224
gini_area_job	-0.332*** 0.000	-0.355*** 0.000	0.122* 0.096	0.117* 0.078
ldensity	-0.413*** 0.000	-0.546*** 0.000	0.488*** 0.000	0.445*** 0.000
gini_pop_job	-0.162 0.115	-0.338*** 0.000	-0.116 0.131	-0.132 0.123
house_age	-0.032 0.739	-0.092 0.308	0.184 0.129	0.129* 0.090
rel_public_comp	-0.057 0.342	-0.067 0.249	-0.115** 0.047	-0.115** 0.037
area	0.595*** 0.000	0.542*** 0.000	0.588*** 0.000	0.578*** 0.000
dummy_centre	-0.021 0.914	-0.050 0.801	-0.285 0.104	-0.306* 0.064
dummy_south	0.537* 0.062	0.705*** 0.007	-0.122 0.659	-0.086 0.705
demo_structure	-0.110 0.396	-0.128 0.321	-0.037 0.715	-0.044 0.712
graduates	0.282*** 0.000	0.289*** 0.000	0.225*** 0.000	0.225*** 0.001
const	-0.167 0.292	-0.212 0.165	0.145 0.355	0.140 0.275
rho	0.267*** 0.008	0.287*** 0.004		
Number of obs.	82	82	82	82
Adj. R-squared	0.745	0.746	0.786	0.789
Log likelihood	-59.09	-58.88	-46.60	-46.18
AIC	146.17	145.77	117.19	116.36
BIC	179.87	179.46	146.07	145.24

Source: Our elaboration on Istat Census data (2001).

* $p < .10$; ** $p < .05$; *** $p < .01$.

Estimation results show that urban compactness is an important factor in explaining commuting costs. A higher density is associated with more sustainable mobility patterns in terms of CO₂ emissions, but with a longer travel duration. The relation between density and commuting time is in accordance with the results of Glaeser and Kahn (2004) and Gordon, Kumar, and Richardson (1989) for U.S. MAs. Following these authors, this result is due, at least in part, to a congestion effect that arises in high-density cities. Hence, a trade-off between external and internal costs comes out and the net effect of density is still unknown.

As regards urban functional diversity, its influence on mobility costs remains unclear. Although the coefficients of the mixed land-use variable (*gini_pop_job*) are negative for the external costs model, they lose statistical significance in one of the two specifications. Thus, it cannot be stressed that functional diversification has a significant relationship with the external costs of commuting. The same results hold for private costs specifications, for which coefficients are always insignificant, though in this case the sign of the relationship is positive.

Urban size is probably the most important factor that is related to the impact of mobility. As was expected, larger size is associated with higher CO₂ emissions and with more time spent on travelling. These results can be explained by the longer distances that on average have to be travelled in larger MAs, which affect both time and pollutant emissions.

As far as the other control variables are concerned, it appears that the age of the housing stock does not have a statistically significant effect. Coefficients of the dummy variables in relation to the MAs' geographic localisation are also insignificant, except for the Southern MAs with respect to the external costs specifications. More specifically, southern cities show a higher impact in terms of CO₂ emissions. This can be due to longer distances or to the scarce use of public means of transport. On the other hand, a higher relative competitiveness of a mass transit system is associated with a shorter duration of commutes, as shown by the coefficient of the *rel_public_comp* variable.

Among socio-demographics, the share of graduates is associated with a higher level of CO₂ emissions and higher travel times as was expected and explained above. This result is stable for the two model specifications that have been estimated. The age of the working population (*demo_structure*), on the other hand, does not seem to play a statistically significant role.

Finally, in the spatial lag model the spatial parameter (*rho*) is positive and significant, showing that a proximity effect exists. In other words, neighbouring MAs have similar characteristics which are not taken on by the model's regressors, but which can be associated to other factors related to proximity (e.g., cultural and technological factors).

Concluding Remarks

Contemporary MAs are characterised by an increasingly polycentric spatial structure. The polycentric paradigm promotes this concept as a planning tool to enhance local competitiveness and sustainability. However, polycentricity is not yet a well-defined concept and its effects on competitiveness and sustainability still needs to be investigated both theoretically and empirically.

The contribution of this work relies firstly on the formal definition of polycentricity at the metropolitan scale and on its quantitative measurement for comparative analysis. Secondly, and in more depth, it sheds some light on the effects of the degree of urban polycentricity on two main components of commuting costs.

Polycentricity has been conceptualised from an interaction perspective and it has been measured with appropriate indicators based on tools of social network analysis. A novel indicator of polycentricity has been introduced, and another one has also been computed for reasons of robustness and completeness of the analysis.

The analysis on 82 Italian MAs shows that polycentricity is a sustainable model of urban spatial structure, especially from an environmental perspective because it is associated with a reduction in CO₂ emissions due to commuting. The results are robust for two different measures of polycentricity, and they are also consistent with the morphological indicators of employment concentration within MAs. However, the degree of polycentricity does not seem to influence the time spent on travelling in a significant way.

Considering other characteristics of urban spatial structure, it comes out that urban dispersion is associated with higher external costs, but also with smaller private costs. Hence, the analysis shows a trade-off between the need to contain collective costs of mobility (such as pollution and greenhouse gases) and the need to satisfy housing preferences for low-density areas/neighbourhoods.

NOTES

1. The z_{ij} element of the matrix represents the link between the i -th municipality and the j -th municipality. The z_{ij} value is 1 for the four-nearest-neighbour municipalities, with a minimum threshold of 5 commuters. In all the other cases the value is 0.
2. Networks are metropolitan areas and nodes are municipalities. Following Green (2007), matrixes contain the number of commuters from municipality i to municipality j , so that they have not been dichotomised.
3. These units of analysis have been identified in Boix and Veneri (2009).
4. Robust LM test of spatial autocorrelation suggested using a spatial lag model, whose residuals do not show spatial autocorrelation anymore, so that coefficients are consistent. Results are also robust to the change of the weight matrix.
5. There are not any problems of residual's spatial auto-correlation for model 2, since the Moran I is not significant, checking these results with different weighting matrixes.

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Appendix

TABLE A1. BI-VARIATE CORRELATIONS BETWEEN INDEPENDENT VARIABLES.

Variables	polyc_green	polyc_nsr	polyc_g_area_job	ldensity	g_pop_job	house_a	rel_pub_c	area	d_centre	d_south	demo	grad.
polyc_green	1.00											
polyc_nsr	0.57	1.00										
gini_area_job	0.33	0.22	1.00									
ldensity	0.25	0.00	-0.17	1.00								
gini_pop_job	0.43	0.06	0.30	-0.27	1.00							
house_age	0.17	-0.02	-0.18	-0.05	0.37	1.00						
rel_public_comp	0.16	0.14	0.03	0.09	0.00	0.01	1.00					
area	0.50	0.32	0.36	0.02	0.13	-0.06	-0.17	1.00				
dummy_centre	-0.32	-0.22	-0.14	-0.17	-0.44	-0.16	0.00	0.04	1.00			
dummy_south	-0.06	-0.20	0.01	0.04	0.55	0.50	-0.05	-0.15	-0.53	1.00		
demo_structure	-0.12	0.09	0.06	-0.12	-0.48	-0.74	0.00	0.08	0.33	-0.78	1.00	
graduates	0.03	-0.10	0.29	0.09	-0.04	-0.07	-0.06	0.26	0.28	-0.09	0.16	1.00

TABLE A2. MULTI-COLLINEARITY TEST.

Variable	VIF	1/VIF
dummy_south	5.62	0.17781
demo_structre	5.47	0.18285
polyc_green	4.11	0.24308
gini_pop_job	3.46	0.28920
house_age	3.03	0.33048
dummy_centre	2.36	0.42392
ldensity	2.28	0.43862
area	1.86	0.53874
gini_area_job	1.70	0.58786
graduates	1.54	0.65049
rel_public_comp	1.16	0.85928
Mean VIF	2.96	

values above ten may indicate a collinearity problem.

TABLE A3. SPATIAL AUTO-CORRELATION TESTS FOR $EIM(A)$ MODEL WITH OLS.

Index	Spat. matrix	I	E(I)	SD(I)	z	p-value*
Moran's I	W1	0.088	-0.012	0.07	1.437	0.075
Moran's I	W2	0.092	-0.012	0.075	1.391	0.082
Geary's c	W1	0.897	1	0.07	-1.466	0.071
Geary's c	W2	0.877	1	0.077	-1.607	0.054

* one-tail test.

W1, a binary contiguity matrix; W2, a four nearest neighbours matrix with the inverse of distances between MAs.

TABLE A4. SPATIAL AUTO-CORRELATION TESTS FOR MODEL $EIM(B)$ MODEL WITH OLS.

Index	Spat. matrix	I	E(I)	SD(I)	z	p-value*
Moran's I	W1	0.14	-0.012	0.07	2.184	0.014
Moran's I	W2	0.151	-0.012	0.075	2.17	0.015
Geary's c	W1	0.856	1	0.07	-2.038	0.021
Geary's c	W2	0.821	1	0.077	-2.333	0.011

* one-tail test.

W1, a binary contiguity matrix; W2, a four nearest neighbours matrix with the inverse of distances between MAs.